

# Bank Customer Churn Analysis

*Data Cleaning, Exploratory Analysis & Business Recommendations*

Tool: Microsoft Excel | Records: 10,000 | Original fields: 13

## 1. Project Overview

This project analyses customer churn in a European bank customer dataset using Microsoft Excel. The objective is to validate the data, identify customer segments associated with different churn patterns, communicate the main findings visually, and translate the findings into practical business recommendations.

## 2. Dataset Overview

Dataset: Bank Customer Churn Dataset

Records: 10,000 customers

Original fields: 13

Tool: Microsoft Excel

Dataset structure: Single table

Source: Maven Analytics Data Playground — Bank Customer Churn. Maven describes the dataset as account information for 10,000 customers at a European bank and lists Kaggle as the source.

Exited is the customer churn outcome: 0 = Customer stayed; 1 = Customer exited.

### Revised Excel Headers

Use these readable headers in the Excel analysis sheet. The original dataset has 13 core fields; Age Group, Tenure Group and Account Balance Group are analysis-created fields.

| # | Excel Header       | Status                       | Notes                    |
|---|--------------------|------------------------------|--------------------------|
| 1 | Customer ID        | Original / Renamed / Created | Readable analysis header |
| 2 | Surname            | Original / Renamed / Created | Readable analysis header |
| 3 | Credit Score       | Original / Renamed / Created | Readable analysis header |
| 4 | Country            | Original / Renamed / Created | Readable analysis header |
| 5 | Gender             | Original / Renamed / Created | Readable analysis header |
| 6 | Age                | Original / Renamed / Created | Readable analysis header |
| 7 | Tenure (Years)     | Original / Renamed / Created | Readable analysis header |
| 8 | Account Balance    | Original / Renamed / Created | Readable analysis header |
| 9 | Number of Products | Original / Renamed / Created | Readable analysis header |

|    |                       |                              |                          |
|----|-----------------------|------------------------------|--------------------------|
| 10 | Has Credit Card       | Original / Renamed / Created | Readable analysis header |
| 11 | Is Active Member      | Original / Renamed / Created | Readable analysis header |
| 12 | Estimated Salary      | Original / Renamed / Created | Readable analysis header |
| 13 | Exited                | Original / Renamed / Created | Readable analysis header |
| 14 | Age Group             | Created                      | Readable analysis header |
| 15 | Tenure Group          | Created                      | Readable analysis header |
| 16 | Account Balance Group | Created                      | Readable analysis header |

### 3. Data Quality Inspection

The dataset was reviewed for missing values, duplicate records, inconsistent categories, invalid ranges and formatting issues before analysis.

| Check              | Result                                                                 | Action                     |
|--------------------|------------------------------------------------------------------------|----------------------------|
| Missing values     | No missing values identified across the 13 original columns.           | No changes required.       |
| Duplicate records  | No duplicate customer IDs or duplicate entire records identified.      | No records removed.        |
| Country            | France, Spain and Germany; no inconsistent or abbreviated names.       | No changes required.       |
| Gender             | Female and Male; no inconsistent labels.                               | No changes required.       |
| Age                | 18–92; no missing or obviously invalid ages.                           | No changes required.       |
| Credit Score       | 350–850; no missing or obviously invalid scores.                       | No changes required.       |
| Tenure             | 0–10 years; no missing or obviously invalid values.                    | No changes required.       |
| Number of Products | Values 1–4; no unexpected values.                                      | No changes required.       |
| Has Credit Card    | Binary values 0 and 1.                                                 | No changes required.       |
| Is Active Member   | Binary values 0 and 1.                                                 | No changes required.       |
| Estimated Salary   | 11.58–199,992.48; no missing values, text or currency symbols.         | No changes required.       |
| Exited             | Binary values 0 and 1.                                                 | No changes required.       |
| Headers            | Original headers reviewed; readable headers applied in analysis sheet. | Use revised headers above. |

## 4. Churn Analysis

### I. Overall Churn

**Finding:** 2,037 of 10,000 customers exited, resulting in a 20.37% churn rate.

**Business implication:** The bank has a meaningful churn base that warrants segment-level investigation.

### II. Churn by Country

**Finding:** France: 16.15%; Spain: 16.67%; Germany: 32.44%. Germany was substantially above the overall churn rate of 20.37%.

**Business implication:** Investigate customer experience, product mix, activity and account characteristics in Germany before designing country-specific retention actions.

### III. Churn by Gender

**Finding:** Female customers had a 25.07% churn rate versus 16.46% among male customers, an 8.61 percentage-point difference.

**Business implication:** Gender is associated with different churn patterns in this dataset. Further segmentation is needed before attributing the difference to gender itself.

### IV. Customer Activity and Churn

**Finding:** Inactive customers had a 26.85% churn rate versus 14.27% among active customers, a 12.58 percentage-point difference.

**Business implication:** Inactive customers are a clear engagement segment for re-engagement and retention analysis.

### V. Churn by Age Group

**Finding:** 18–29: 7.56%; 30–39: 10.88%; 40–49: 30.79%; 50–59: 56.04%; 60+: 27.95%.

**Business implication:** Customers aged 40–59, especially 50–59, should receive further investigation to understand the drivers of higher churn.

### VI. Products and Age

**Finding:** Two-product customers had notably lower churn than one-product customers. Three-product churn was much higher, and all observed four-product customers had exited.

**Business implication:** Three- and four-product groups were much smaller, so the extreme rates should be treated as signals for further investigation, not proof that more products cause churn.

### VII. Average Account Balance

**Finding:** Average balance was 72,745.30 for customers who stayed and 91,108.54 for customers who exited, a difference of approximately 18,363.24.

**Business implication:** Higher-balance customers may represent a potentially valuable segment at risk of churn.

## VIII. Credit Score

**Finding:** Average credit score was 651.85 for customers who stayed and 645.35 for customers who exited, a difference of 6.50 points.

**Business implication:** Credit score showed only a modest difference and appears less prominent than several other characteristics examined.

## IX. Churn by Balance Range

**Finding:** 0: 13.82%; 1–50,000: 34.67%; 50,001–100,000: 19.88%; 100,001–150,000: 25.77%; 150,001+: 23.12%. The 1–50,000 group contained only 75 customers.

**Business implication:** Balance is associated with different churn patterns, but the relationship is not simply linear. Small segments should be interpreted cautiously.

Formula used for exited customers: =COUNTIF(M2:M10001,1)

Formula used for churn rate: =COUNTIF(M2:M10001,1)/COUNTA(A2:A10001)

## 5. Key Findings

| Finding                 | Evidence                                                                                                                    |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Overall churn           | 2,037 of 10,000 customers exited, producing a 20.37% churn rate.                                                            |
| Geography               | Germany had the highest churn at 32.44%, versus 16.15% in France and 16.67% in Spain.                                       |
| Customer activity       | Inactive customers had 26.85% churn versus 14.27% among active customers.                                                   |
| Age                     | Churn increased considerably with age, reaching 56.04% among customers aged 50–59.                                          |
| Gender                  | Female customers had higher churn than male customers: 25.07% versus 16.46%.                                                |
| Products                | Two-product customers had 7.58% churn; three- and four-product groups showed very high rates but were much smaller.         |
| Account balance         | Exited customers had a higher average balance than retained customers; balance ranges also showed different churn patterns. |
| Credit score and tenure | Average credit scores were similar, and tenure churn ranged from 18.87% to 21.30%.                                          |

## 6. Visualizations

Six charts were created in Microsoft Excel:

1. Overall Customer Churn
2. Customer Churn Rate by Country
3. Customer Churn Rate by Age Group
4. Customer Churn Rate by Activity Status
5. Customer Churn Rate by Number of Products
6. Customer Churn Rate by Account Balance

The completed charts remain in the Excel workbook and can be inserted into this section for a visual-report version.

## 7. Business Recommendations

| Priority Area          | Evidence from Analysis                                                                                | Business Recommendation                                                                                                                         |
|------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer Engagement    | Inactive: 26.85% churn vs Active: 14.27%.                                                             | Develop targeted re-engagement campaigns, personalised offers, product education, reminders and digital engagement initiatives.                 |
| Germany                | Germany: 32.44% vs France: 16.15% and Spain: 16.67%.                                                  | Investigate customer experience, product mix and service patterns in Germany and develop evidence-based retention actions.                      |
| Older Customers        | Age 50–59: 56.04%; age 40–49: 30.79%.                                                                 | Prioritise retention analysis for customers aged 40–59 and investigate differences in needs, product suitability and service expectations.      |
| High-Value Customers   | Exited average balance: 91,108.54 vs stayed: 72,745.30.                                               | Include account value when prioritising retention efforts so potentially valuable at-risk customers receive appropriate attention.              |
| Product Usage          | Two-product churn: 7.58%; three- and four-product groups had very high rates but were smaller.        | Investigate whether multi-product customers experience complexity, dissatisfaction or service issues. Do not assume product count causes churn. |
| Balance Segments       | 1–50,000 had 34.67% churn but only 75 customers; 100,001–150,000 had 25.77%.                          | Monitor balance segments while avoiding major decisions based on very small groups.                                                             |
| Lower-Priority Factors | Tenure, credit-card ownership, credit score and estimated salary showed relatively small differences. | Keep these factors in the monitoring framework but give them lower immediate priority unless further analysis identifies stronger interactions. |

## 8. Executive Summary

This project analysed customer churn within a dataset of 10,000 customers from a European bank. The objective was to identify customer characteristics and segments associated with higher churn and translate the findings into potential retention priorities. The analysis was conducted using Microsoft Excel and included data validation, descriptive analysis, segmentation and visualisation.

The dataset contained 13 original customer-related fields, including credit score, geography, gender, age, tenure, account balance, number of products, credit-card ownership, active membership status, estimated salary and customer exit status. Data quality checks found no missing values or duplicate records, and categorical and numerical fields were reviewed for consistency.

Overall, 2,037 customers had exited the bank, resulting in a churn rate of 20.37%.

Germany had the highest geographic churn rate at 32.44%, compared with 16.15% in France and 16.67% in Spain. Inactive customers had a churn rate of 26.85%, compared with 14.27% among active customers.

Age was another prominent pattern. Churn reached 56.04% among customers aged 50–59, while customers aged 40–49 had 30.79% churn and customers aged 18–29 had 7.56%.

Exited customers had a higher average account balance than customers who stayed, at 91,108.54 compared with 72,745.30. Balance-range analysis also showed different churn patterns, although smaller groups require cautious interpretation.

Product usage produced an unusual pattern. Customers with two products had the lowest observed churn rate at 7.58%, while customers with three or four products showed substantially higher rates. Because those groups were relatively small, the findings should be treated as signals for further investigation rather than definitive conclusions.

Tenure, credit-card ownership, credit score and estimated salary showed relatively limited differences in churn and therefore appear to be lower-priority factors based on this descriptive analysis.

Based on these findings, the bank could prioritise customer engagement initiatives for inactive customers, investigate elevated churn in Germany, develop targeted retention strategies for customers aged 40–59, and incorporate customer account value into retention prioritisation. Further analysis would be required to determine causal drivers and the strongest predictors of churn.

## 9. Limitations

This analysis is descriptive and identifies patterns and associations within the customer data; it does not establish causal relationships. Some segments, particularly customers with three or four products and customers with balances between 1 and 50,000, contain relatively few observations and should therefore be interpreted cautiously. Further analysis using statistical testing or predictive modelling would be required to determine which factors are the strongest predictors of customer churn.

## 10. Conclusion

The analysis identified customer activity, geography, age, account balance and product usage as important areas for further investigation of customer churn. The findings provide a basis for targeted retention analysis while recognising that additional statistical or predictive analysis would be required to determine the strongest predictors of churn.

## Dataset Source

Maven Analytics Data Playground — Bank Customer Churn. Original source listed by Maven Analytics: Kaggle. Maven describes the dataset as account information for 10,000 customers at a European bank with 13 fields.